

Tahsin Tajwar Tanni

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 Portfolio |  LinkedIn |  GitHub |  Google Scholar

Profile

I'm a Computer Science and Engineering graduate from BRAC University with experience spanning AI automation, machine learning research, and full-stack web development and a published paper on language model security. I build LLM-powered products end to end, from RAG pipelines and agentic workflows to a live SaaS platform, and bring a research habit of measuring what a system actually does before trusting it.

Skills

- **AI and LLM Engineering:** OpenAI API, Anthropic API, RAG, agentic workflows, prompt engineering, n8n, Hugging Face Spaces
- **Machine Learning:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, Matplotlib, Seaborn; model fine-tuning, adversarial ML, benchmarking
- **Languages:** Python, JavaScript, TypeScript, C, Solidity
- **Web and Backend:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, Prisma ORM, Clerk, Supabase
- **Databases and Tools:** PostgreSQL, MySQL, MongoDB, SQLite; Git, GitHub, Docker, Postman, Vercel, Cisco Packet Tracer

Experience

- AI and Automation Engineer**, 180 RE, Toronto, Canada (remote) May 2026 – Present
- Build LLM-powered automation workflows on the OpenAI and Anthropic APIs, automating multi-step business processes through n8n and third-party integrations.
 - Develop and deploy AI chatbots and RAG pipelines into client websites to automate user interactions and support.
 - Apply prompt engineering to tune LLM outputs for accuracy and safety across use cases.
- Python Tutor**, Zentorra Feb 2025 – May 2025
- Tutored high school students in Python from fundamentals to advanced topics and guided their coding projects.
- Junior Web Developer (Part-time)**, Weabers, Chattogram Feb 2023 – Jun 2023
- Built and maintained responsive web applications, reusable UI components and API integrations; debugged and shipped feature updates to production.
- Junior Executive, IT Department**, Robotics Club of BRAC University Jun 2022 – Aug 2023
- Managed the club's internal digital platforms and technical resources with the IT team.

Projects

- ScopeVanta**  *Next.js, TypeScript, Anthropic API, PostgreSQL, Prisma, Clerk, Supabase, Vercel*
- Live SaaS platform for agencies and consultants that turns a client brief into a structured, priced and risk-scored proposal; includes margin modeling with floor, recommended and premium scenarios, proposal auditing, client deal rooms, scope baselines and automated change orders.
 - Shipped with three subscription tiers and a free trial; built the full stack from AI analysis to billing and client portal.
- RAG-Slackbot**  *n8n, Supabase, Ollama, Google Gemini, Slack API*
- Slack-integrated RAG chatbot for real-time querying of internal PDF knowledge bases; automated ingestion, chunking and vector storage with context-aware answers, confidence-based escalation and source attribution.
- ProfMail: AI-Powered Academic Cold Email Generator**  *React, TypeScript, Vite, Tailwind CSS*
- Web app that generates personalized cold emails for graduate applicants, with live professor lab scraping, research-overlap analysis against the applicant's thesis, tone and length controls, and an editing workspace.
- AI Professor Finder**  *n8n, SerpAPI, Google Scholar, Gemini, Google Sheets*
- Automated workflow that discovers MSc/PhD supervisors, pulls their recent publications, compares them against a thesis, and outputs compatibility scores and funding recommendations.
- VAE-DEC Clustering with Uncertainty**  *PyTorch, scikit-learn, pandas*
- Variational Autoencoder combined with Deep Embedded Clustering for unsupervised image clustering, with entropy and latent-variance uncertainty estimates to evaluate clustering confidence.

ThesisFlow: Thesis Management System [↗](#)

MongoDB, Express.js, React, Node.js

- Web platform that centralizes thesis submission, supervision and tracking with role-based dashboards for students and supervisors, structured feedback, progress tracking and AI-powered academic guidance; 2nd Runner-up, CSE471 Project Demonstration, BRAC University.

Realtime Collaborative Playlist [↗](#)

Next.js, React, TypeScript, Prisma, SQLite, Tailwind CSS, Framer Motion

- Multi-user playlist app with live synchronization over Server-Sent Events, drag-and-drop reordering using fractional indexing, a voting system and now-playing simulation.

Stock Forecasting Analysis Toolkit [↗](#)

pandas, statsmodels, Prophet, scikit-learn, TensorFlow, Gradio

- Interactive Hugging Face Space for time-series forecasting and synthetic data generation, letting users run and compare models and visualize results through a web interface.

Research

Student Researcher, Elite Research Lab LLC

Sep 2026 – Present

- Lead a four-person team on a turn-level detection framework for unhealthy relationship formation in mental health chatbot conversations; proposal submitted to the lab's Research Training and Competition 2026.

Undergraduate Researcher, BRAC University

Jan 2025 – Jan 2026

- Built the evaluation dataset and implemented the defense for **Anchor-Guided Repair**, a fine-tuning method that repairs language models damaged by weight noise and low-precision quantization, recovering 60 to 80 percent of the induced loss across FP16, INT8 and 4-bit precision.
- **Publication:** Rohan, Khan, **Tanni**, Fardin, Bushra (2026). Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks. *IndabaX Nigeria 2026*, PMLR 319, 368–381. [↗](#)

Phantom Guard: slopsquatting and package hallucination study

Independent

- Built a corpus of about 400 real malicious package incidents from GitHub Security Advisories and OSV.dev and found a reproducible, ecosystem-stratified signal linking LLM hallucination-style naming to payload sophistication.
- Caught and corrected a construct-validity error by rebuilding the labeling taxonomy from the data; confirmed the result on two independent pulls using Fisher's exact and Cochran-Mantel-Haenszel tests.

Education

BRAC University, BSc in Computer Science and Engineering

2022 – Feb 2026

- CGPA: 3.60 / 4.00. Vice Chancellor's List (Summer 2025), Dean's List (Fall 2024), Academic Distinction (2026)

Govt. Haji Muhammad Mohsin College, Higher Secondary Certificate (HSC)

2018 – 2020

- GPA: 5.00 / 5.00

Bangladesh Mohila Samity Girls' High School and College, Secondary School Certificate (SSC)

2004 – 2018

- GPA: 5.00 / 5.00

Achievements and Certifications

- **2nd Runner-up**, CSE471 Project Demonstration, BRAC University, with ThesisFlow, a Thesis Management System featuring role-based dashboards, progress tracking, structured feedback and AI-powered academic guidance.
- **AI Prompt Engineer (AI CERTs)** via AgentX Bangladesh (Netcom Learning and Microsoft): advanced prompt design (zero and few shot, chain of thought), PFSET, PAIPS, IPS, XML steering, agent workflows and ethical AI use.
- **Intro to Machine Learning**, Kaggle: foundations of data science and machine learning with practical applications.
- **Intermediate Web Development**, BRAC University Residential Semester: practical skills in designing responsive websites.